

Lecture 7-9: Portfolio Optimization

Instructor: Zijie Zhou, IEDA, HKUST

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Outline

- 1 Fundamentals
- 2 Portfolio constraints
- 3 Performance measures
- 4 Heuristic portfolios
- 5 Risk-based portfolios
- 6 Summary

- **Asset prices and returns:**

- Denote prices of N assets at time t as $\mathbf{p}_t \in \mathbb{R}^N$.
- Linear returns defined as

$$\mathbf{r}_t^{\text{lin}} = \frac{\mathbf{p}_t}{\mathbf{p}_{t-1}} - \mathbf{1}.$$

- Log-returns defined as

$$\mathbf{r}_t^{\text{log}} = \log \mathbf{p}_t - \log \mathbf{p}_{t-1}.$$

- **Modeling objectives:**

- Aim to forecast returns at time t using historical data up to $t - 1$ ($\mathcal{F}_{t-1}$).
- Focus on conditional first- and second-order moments (mean vector and covariance matrix).

- **Mean vector and covariance matrix:**

- Conditional mean vector for log-returns: $\boldsymbol{\mu}_t^{\text{log}} = \mathbb{E} \left[\mathbf{r}_t^{\text{log}} \mid \mathcal{F}_{t-1} \right]$.
- Conditional covariance matrix for log-returns: $\boldsymbol{\Sigma}_t^{\text{log}} = \text{Cov} \left[\mathbf{r}_t^{\text{log}} \mid \mathcal{F}_{t-1} \right]$.

- **Preference for log-returns:**

- Financial models often prefer log-returns for modeling, despite needing linear returns for portfolio performance.
- Linear and log-returns are approximated as similar for small return values.

- **Approximations between linear and log-returns:**

- Linear mean and covariance approximated by log counterparts: $\mu_t^{\text{lin}} \approx \mu_t^{\text{log}}$, $\Sigma_t^{\text{lin}} \approx \Sigma_t^{\text{log}}$.
- Exact mathematical relationships allow deriving linear moments from log moments.

- **Simplification for econometric models:**

- Time dependency of moments often dropped for simplicity.
- Linear moments (μ , Σ) referred to by default, especially under i.i.d. model assumption with no time dependency.

- **Portfolio definition:**

- Allocation of budget among N risky assets, with uninvested amount as cash.
- Represented by capital allocation $\mathbf{w}_t^{\text{cap}} \in \mathbb{R}^N$, where $w_{i,t}^{\text{cap}}$ is the dollar amount for asset i .
- Cash represented separately as $c_t^{\text{cap}} \in \mathbb{R}$ or included in $\mathbf{w}_t^{\text{cap}}$ as a riskless asset.
- Portfolio can also be represented by units held $\mathbf{w}_t^{\text{units}} \in \mathbb{R}^N$ (e.g., shares or coins).

- **Dollar amount variation:**

- Keeping unit amount constant, dollar amount changes with asset prices:

$$\mathbf{w}_t^{\text{cap}} = \mathbf{w}_t^{\text{units}} \odot \mathbf{p}_t.$$

- Portfolio time evolution:

$$\mathbf{w}_t^{\text{cap}} = \mathbf{w}_{t-1}^{\text{cap}} \odot (\mathbf{1} + \mathbf{r}_t^{\text{lin}}).$$

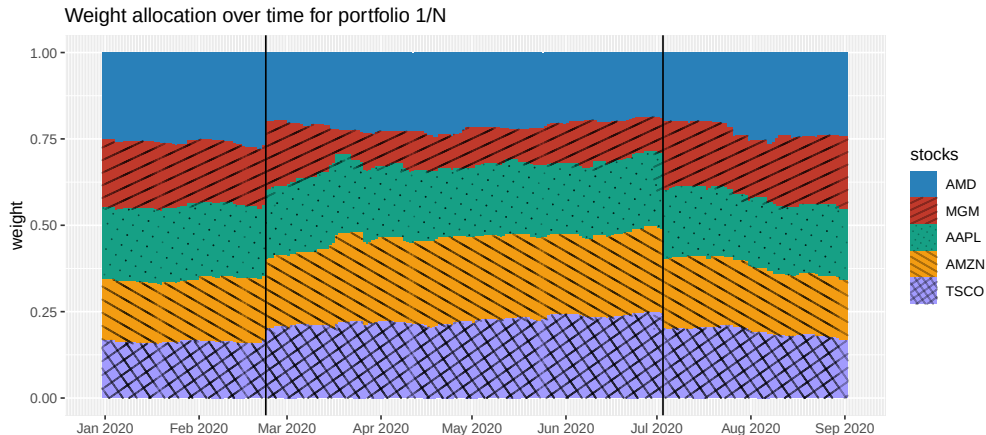
- Cash remains constant unless adjusted for contributions or withdrawals.

- **Rebalancing for fixed dollar amount:**

- To maintain a fixed dollar allocation, portfolios require regular rebalancing.
- Rebalancing adjusts $\mathbf{w}_t^{\text{cap}}$ to original positions, incurring transaction costs.

Portfolio rebalancing

Evolution of the $1/N$ portfolio over time with effect of rebalancing (vertical lines):



- **Portfolio NAV definition:**

- The portfolio *net asset value* (NAV) at time t includes current market valuation and cash:

$$\text{NAV}_t \triangleq \mathbf{1}^T \mathbf{w}_t^{\text{cap}} + c_t^{\text{cap}}.$$

- **NAV evolution:**

- Evolution derived from portfolio time evolution:

$$\text{NAV}_t = \text{NAV}_{t-1} + (\mathbf{w}_{t-1}^{\text{cap}})^T \mathbf{r}_t^{\text{lin}}.$$

- Indicates NAV change depends on assets' returns.

- **Portfolio return:**

- Defined as the change in NAV relative to previous NAV:

$$R_t^{\text{portf}} \triangleq \frac{\text{NAV}_t - \text{NAV}_{t-1}}{\text{NAV}_{t-1}} = \mathbf{w}_{t-1}^T \mathbf{r}_t^{\text{lin}}.$$

- **Normalized portfolio:**

- Portfolio normalized with respect to current NAV:

$$\mathbf{w}_t = \mathbf{w}_t^{\text{cap}} / \text{NAV}_t$$

- Used in portfolio design and optimization.

- **Asset additivity property:**

- Linear returns exhibit asset additivity, crucial for portfolio return calculation.

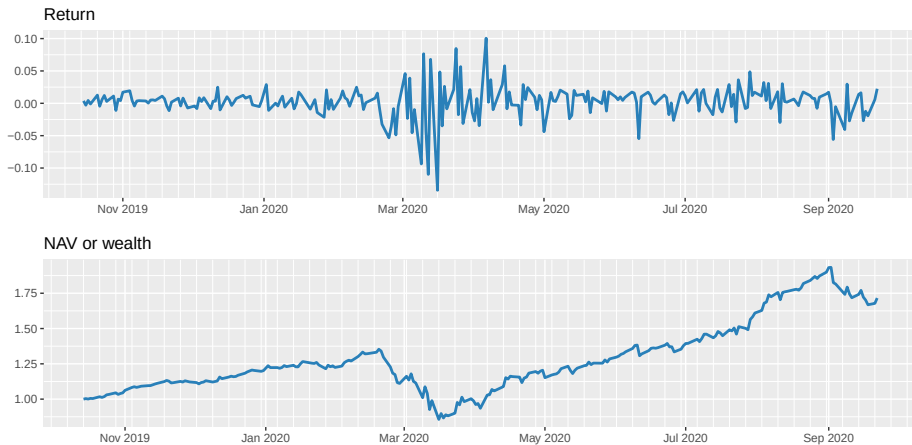
- **Adjustment for cash contributions:**

- If cash contributions occur, adjust portfolio return calculation:

$$R_t^{\text{portf}} \triangleq \frac{\text{NAV}_t - \text{NAV}_{t-1} - \text{contribution}_{t-1}}{\text{NAV}_{t-1}} = \mathbf{w}_{t-1}^T \mathbf{r}_t^{\text{lin}}.$$

NAV and return

Return and NAV of the $1/N$ portfolio over time:



- **Cumulative return and NAV:**

- Cumulative return: $NAV_t/NAV_0 - 1$.
- Does not include contributions or redemptions.
- Cumulative return plot starts at zero; NAV plot starts at 1 (if normalized).

- **Importance of position size:**

- Cumulative return and NAV depend on portfolio design and invested budget.
- Budget may include cash reserves.
- Capital invested is referred to as *position size*.

Cumulative return and sizing

- **Full reinvesting:**

- Normalized portfolio $\mathbf{w}_t$ with full reinvestment of current NAV: $\mathbf{w}_t^{\text{cap}} = \text{NAV}_t \times \mathbf{w}_t$.
- NAV evolution:

$$\text{NAV}_t = \text{NAV}_{t-1} \times (1 + R_t^{\text{portf}})$$

- Geometric growth:

$$\text{NAV}_t = \text{NAV}_0 \times (1 + R_1^{\text{portf}}) \times \dots \times (1 + R_t^{\text{portf}})$$

- **Constant reinvesting:**

- Normalized portfolio $\mathbf{w}_t$ with constant reinvestment of initial NAV: $\mathbf{w}_t^{\text{cap}} = \text{NAV}_0 \times \mathbf{w}_t$.
- NAV evolution:

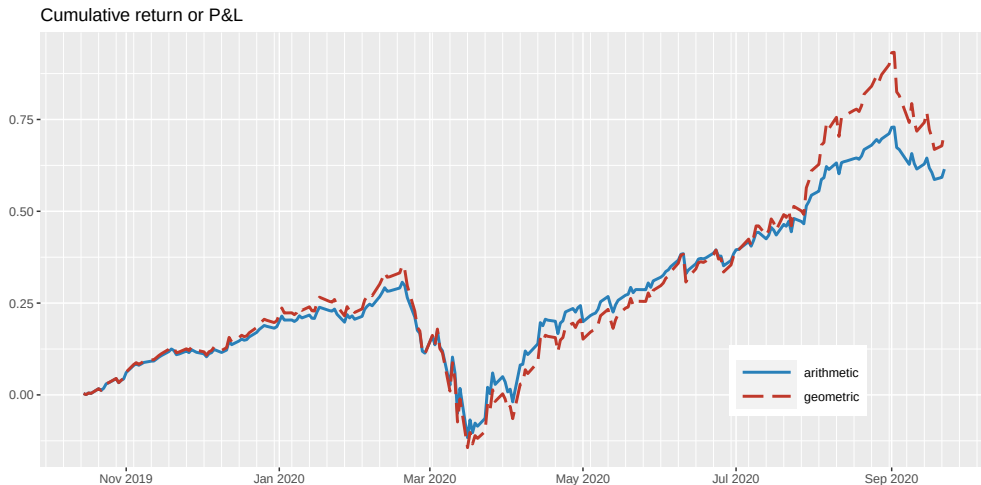
$$\text{NAV}_t = \text{NAV}_{t-1} + \text{NAV}_0 \times R_t^{\text{portf}}$$

- Arithmetic growth:

$$\text{NAV}_t = \text{NAV}_0 \times (1 + R_1^{\text{portf}} + R_2^{\text{portf}} + \dots + R_t^{\text{portf}})$$

Cumulative return

Comparison of arithmetic and geometric cumulative returns:

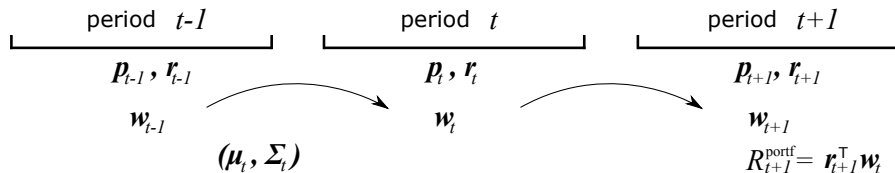


Portfolio notation over time*

- **Portfolio time notation:**

- Portfolio at time t is denoted as $\mathbf{w}_t$.
- $\mathbf{w}_t$ uses information up to time $t - 1$ (moments μ_t and Σ_t).
- Executed at time t , assuming instantaneous execution.

- Portfolio notation over time periods:



Portfolio notation over time*

- **Incorrect return calculation:**

- Multiplying $\mathbf{w}_t$ with return at the same period $\mathbf{r}_t^{\text{lin}}$ is incorrect.
- This would incur look-ahead bias (refer to the topic of backtesting and its dangers).

- **Correct return calculation:**

- $\mathbf{w}_t$ is the portfolio executed and held at time t , with prices $\mathbf{p}_t$.
- Its return can be computed upon observing prices $\mathbf{p}_{t+1}$.
- Correct return calculation: $\mathbf{w}_t^T \mathbf{r}_{t+1}^{\text{lin}}$.

- **Alternative time index definition:**

- Some authors define returns with a shift in the time index.
- This allows $\mathbf{p}_t$ to be multiplied with $\mathbf{r}_t^{\text{lin}}$.
- Care must be taken to maintain consistency in notation and avoid look-ahead bias.

- **First- and second-order moments in portfolio optimization:**

- Expected value and variance are crucial for making informed investment decisions.
- Optimization techniques aim to find portfolios with desirable expected returns and risk levels.

- **Expected value of portfolio return:**

- Given by:

$$\mathbb{E} \left[R_t^{\text{portf}} \right] = \mathbf{w}_{t-1}^T \boldsymbol{\mu}$$

- Represents the mean return of the portfolio.

- **Variance of portfolio return:**

- Given by:

$$\text{Var} \left[R_t^{\text{portf}} \right] = \mathbf{w}_{t-1}^T \boldsymbol{\Sigma} \mathbf{w}_{t-1}$$

- Represents the risk or volatility of the portfolio return.

- **Transaction costs overview:**

- Consist of commission fee and slippage.
- Commission fee: Payment to broker, varies by country and broker.
- Slippage: Difference between expected and actual execution price, affects liquid and illiquid assets differently.

- **Impact on portfolio NAV:**

- Rebalancing to $\mathbf{w}_t^{\text{cap, reb}}$ decreases NAV by transaction cost $\phi(\mathbf{w}_t^{\text{cap}} \rightarrow \mathbf{w}_t^{\text{cap, reb}})$.
- NAV after rebalancing and price change:

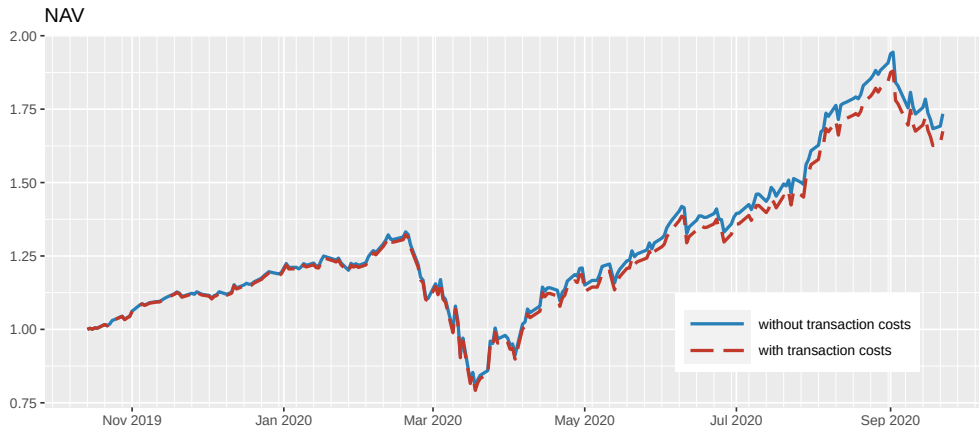
$$\text{NAV}_t = \text{NAV}_{t-1} + (\mathbf{w}_{t-1}^{\text{cap, reb}})^T \mathbf{r}_t^{\text{lin}} - \phi(\mathbf{w}_{t-1}^{\text{cap}} \rightarrow \mathbf{w}_{t-1}^{\text{cap, reb}}).$$

- Portfolio return with transaction costs:

$$R_t^{\text{portf}} = (\mathbf{w}_{t-1}^{\text{reb}})^T \mathbf{r}_t^{\text{lin}} - \phi(\mathbf{w}_{t-1} \rightarrow \mathbf{w}_{t-1}^{\text{reb}}).$$

Transaction costs

Effect of transaction costs on the NAV (with daily rebalancing and 90 bps of fees):



- **Expected value and variance with transaction costs:**

- Expected value:

$$\mathbb{E} \left[R_t^{\text{portf}} \right] = (\mathbf{w}_{t-1}^{\text{reb}})^T \boldsymbol{\mu} - \phi (\mathbf{w}_{t-1} \rightarrow \mathbf{w}_{t-1}^{\text{reb}}) .$$

- Variance:

$$\text{Var} \left[R_t^{\text{portf}} \right] = (\mathbf{w}_{t-1}^{\text{reb}})^T \boldsymbol{\Sigma} \mathbf{w}_{t-1}^{\text{reb}} .$$

Transaction costs*

- **Transaction cost function:**

- Decomposed into fees and slippage:

$$\phi(\mathbf{w}_t \rightarrow \mathbf{w}_t^{\text{reb}}) = \phi^{\text{fees}}(\mathbf{w}_t \rightarrow \mathbf{w}_t^{\text{reb}}) + \phi^{\text{slippage}}(\mathbf{w}_t \rightarrow \mathbf{w}_t^{\text{reb}}).$$

- Fees proportional to turnover:

$$\phi^{\text{fees}}(\mathbf{w}_t \rightarrow \mathbf{w}_t^{\text{reb}}) \approx \tau^{\text{fee}} \|\mathbf{w}_t^{\text{reb}} - \mathbf{w}_t\|_1.$$

- Slippage varies by asset:

$$\phi^{\text{slippage}}(\mathbf{w}_t \rightarrow \mathbf{w}_t^{\text{reb}}) \approx \sum_{i=1}^N \tau_i^{\text{slippage}} |w_{it}^{\text{reb}} - w_{it}|.$$

- **Practical considerations:**

- Account for transaction costs in backtests for realistic performance assessment.
- Monitor strategy turnover and return on turnover due to transaction cost implications.

- **Portfolio rebalancing trade-off:**

- Frequent rebalancing aligns the market portfolio with the desired one.
- Less frequent rebalancing reduces transaction costs.

- **Rebalancing schemes:**

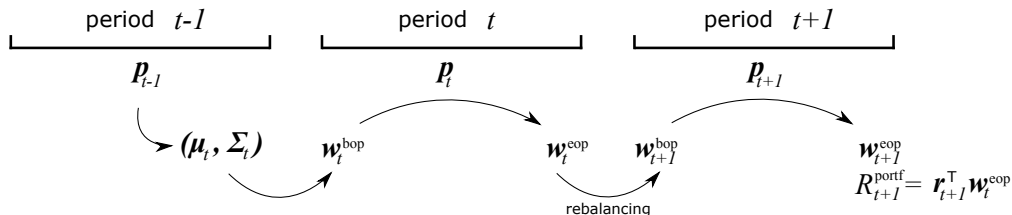
- Calendar-based: Daily, weekly, or monthly rebalancing.
- Adaptive: Rebalance only if portfolio distance exceeds a threshold.
- Measures of difference: Turnover, Mahalanobis distance, tracking error squared distance.

- **Turnover considerations:**

- Turnover is often upper-bounded or penalized to avoid excessive rebalancing.

Portfolio rebalancing*

Portfolio notation over time periods with rebalancing:



- **Time-index notation for rebalancing:**

- Period t divided into beginning of period (bop) and end of period (eop).
- Portfolio w_t separated into w_t^{bop} and w_t^{eop} .

Portfolio rebalancing*

- **Price change and rebalancing:**

- Price change:

$$\mathbf{w}_t^{\text{eop}} = \mathbf{w}_t^{\text{bop}} \odot (\mathbf{1} + \mathbf{r}_t^{\text{lin}}).$$

- Rebalancing incurs transaction costs; if none, then $\mathbf{w}_{t+1}^{\text{bop}} = \mathbf{w}_t^{\text{eop}}.$

- **Fixed portfolio rebalancing:**

- After rebalancing (without transaction costs):

$$\mathbf{w}_t^{\text{bop}} = \mathbf{w}$$

- After price change:

$$\mathbf{w}_t^{\text{eop}} = \mathbf{w} \odot (\mathbf{1} + \mathbf{r}_t^{\text{lin}}).$$

- Portfolio return:

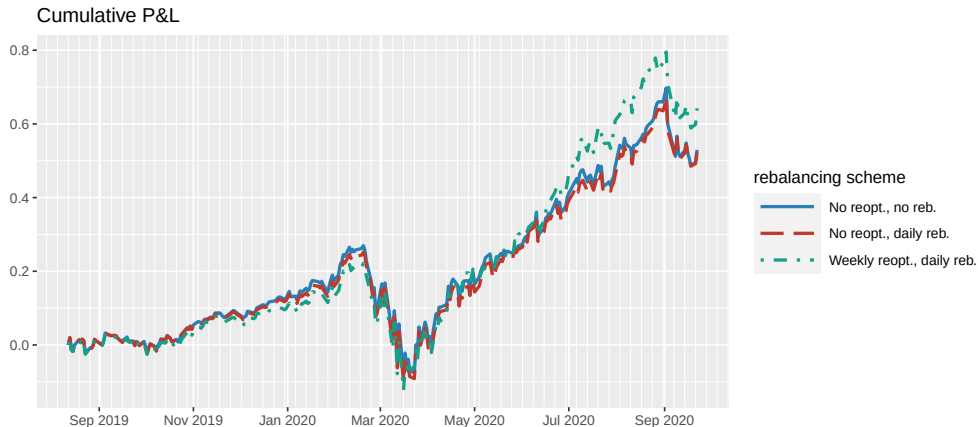
$$R_t^{\text{portf}} = \mathbf{w}^T \mathbf{r}_t^{\text{lin}}.$$

- **Effect of reoptimization and rebalancing:**

- Different schemes affect performance, especially with transaction costs.
- Rebalancing vs. Reoptimization: Execution in the market vs. updating portfolio design based on new data.

Portfolio rebalancing*

Backtest cumulative P&L of a portfolio with different reoptimization and rebalancing schemes:



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Portfolio constraints

- **Common portfolio constraints:**

- Imposed by regulators or brokers: Shorting constraints, leverage constraints, margin requirements.
- Optional (investor's views): Market neutrality, portfolio sparsity, diversity enforcement.
- Basic reference: (Palomar 2025, chap. 6).

- **Examples:**

- long-only or no-shorting constraint
- capital budget constraint
- holding constraints
- cardinality constraint
- turnover constraint
- market-neutral constraint
- dollar-neutral constraint
- diversification constraint
- leverage constraint
- margin requirements

- **Long-only or no-shorting constraint:**

- Investors typically take long positions.
- Short selling allowed in some markets, implying negative elements in $\mathbf{w}$.
- Constraint for no shorting:

$$\mathbf{w} \geq \mathbf{0},$$

which is linear and convex.

- **Capital budget constraint:**

- Without shorting or leverage:

$$\mathbf{1}^T \mathbf{w} + c = 1,$$

linear and convex.

- Without cash variable:

$$\mathbf{1}^T \mathbf{w} \leq 1.$$

- If equality is imposed, $\mathbf{1}^T \mathbf{w} = 1$, the implication is that the cash is zero and the portfolio is fully invested in the risky assets.
- Implies full investment in risky assets if equality is imposed (cash is zero).

- **Holding constraints:**

- Limits on maximum positions for diversification
- May include minimum positions for desired asset holdings.
- Holding constraints are imposed via lower and upper bounds:

$$l \leq \mathbf{w} \leq \mathbf{u},$$

which is linear and convex.

- **Cardinality constraint:**

- Limits the number of active positions to simplify operations:

$$\|\mathbf{w}\|_0 \leq K,$$

nonconvex and challenging to handle.

- **Turnover constraint:**

- Controls portfolio turnover to manage transaction costs:

$$\|\mathbf{w} - \mathbf{w}_0\|_1 \leq u,$$

convex as norms are convex functions.

Portfolio constraints

- **Market-neutral constraint:**

- Avoids market exposure, ensuring portfolio is orthogonal to market “beta”:

$$\beta^T \mathbf{w} = 0,$$

linear and convex.

- **Dollar-neutral constraint:**

- Balances long and short positions:

$$\mathbf{1}^T \mathbf{w} = 0.$$

- Should be used with a leverage constraint due to practical limitations on short selling.

- **Diversification constraint:**

- Promotes allocation across multiple assets:

$$\|\mathbf{w}\|_2^2 \leq D,$$

where $1/N \leq D < 1$.

- Uses the Herfindahl index as a diversification measure, convex constraint.

- **Leverage constraint:**

- Limits borrowing for long and short positions
- The ℓ_1 -norm measures the total amount of long and short positions:

$$\|\mathbf{w}\|_1 \leq L,$$

where L indicates the leverage.

- **Margin requirements:**

- Brokers impose margin requirements for borrowing:

$$m^{\text{long}} \times \mathbf{1}^T \mathbf{w}^+ + m^{\text{short}} \times \mathbf{1}^T \mathbf{w}^- \leq 1,$$

- Separates portfolio into longs ($\mathbf{w}^+$) and shorts ($\mathbf{w}^-$) with specific margin requirements for each.

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- **Portfolio performance assessment:**

- Recall the expression of the portfolio return:

$$R_t^{\text{portf}} = \mathbf{w}^T \mathbf{r}_t$$

- All the performance measures are derived from the portfolio return.
- References: (Bacon 2008), (Palomar 2025, chap. 6).

- **Examples:**

- expected return
- volatility
- Sharpe ratio and information ratio
- downside risk and semivariance
- gain-loss ratio
- Sortino ratio
- Value-at-risk (VaR) and Conditional value-at-risk (CVaR)
- drawdown
- Calmar ratio and Sterling ratio

Performance measures: Expected return

- **Expected return:**

- One-period expected return:

$$\mathbb{E} \left[R_t^{\text{portf}} \right] = \mathbf{w}^T \boldsymbol{\mu}.$$

- Reflects average investment benefit, excluding risk.

- **Annualized return:**

- Scaled by number of periods per year, γ :

$$\text{Annualized Return} = \gamma \times \mathbf{w}^T \boldsymbol{\mu}.$$

- For daily prices: $\gamma = 252$ (stock market) or $\gamma = 365$ (cryptocurrency market).
- For hourly prices: $\gamma = 8,760$ (cryptocurrency market).

Performance measures: Expected return

- **Sample mean annualized return:**

- Arithmetic chaining (assuming T returns):

$$\frac{\gamma}{T} \sum_{t=1}^T R_t^{\text{portf}}.$$

- **Compound annual growth rate (CAGR):**

- Geometric chaining for compounding:

$$\left(\prod_{t=1}^T (1 + R_t^{\text{portf}}) \right)^{\gamma/T} - 1.$$

- Geometric mean is lower than arithmetic mean before annualization.

Performance measures: Volatility

- **Volatility:**

- Standard deviation of portfolio return:

$$\text{Std} \left[R_t^{\text{portf}} \right] = \sqrt{\mathbf{w}^T \Sigma \mathbf{w}}.$$

- Simplest risk measure, variance used by Markowitz.

- **Annualized volatility:**

- Scaled by $\sqrt{\gamma}$:

$$\text{Annualized Volatility} = \sqrt{\gamma} \times \sqrt{\mathbf{w}^T \Sigma \mathbf{w}}.$$

Performance measures: Volatility-adjusted returns

- **Volatility-adjusted returns:**

- Adjusts returns for volatility comparison:

$$\bar{R}_t^{\text{portf}} = \mathbf{w}^T \mathbf{r}_t \times \frac{\sigma^{\text{target}}}{\sigma_t},$$

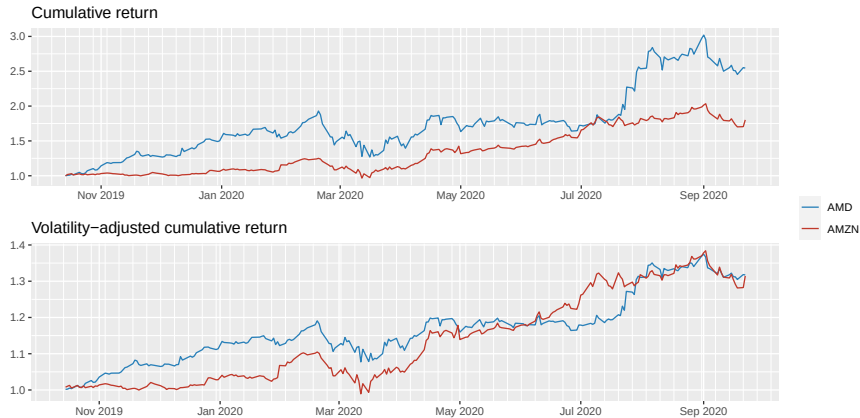
- σ_t : Volatility of $\mathbf{w}^T \mathbf{r}_t$.
- σ^{target} : Desired target volatility.

- **Cumulative returns comparison:**

- Non-adjusted may mislead due to different volatilities.
- Volatility-adjusted provides a better return-volatility trade-off.
- Sharpe ratio captures this trade-off.

Performance measures: Volatility-adjusted returns

Cumulative returns and volatility-adjusted cumulative returns:



Performance measures: Sharpe ratio

- **Sharpe ratio (SR):**

- Risk-adjusted expected excess return:

$$SR = \frac{\mathbf{w}^T \boldsymbol{\mu} - r_f}{\sqrt{\mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}}},$$

- r_f : risk-free rate.
- Measures return per unit of risk.

- **Annualized Sharpe ratio:**

- Adjusted for number of periods per year:

$$\text{Annualized SR} = \sqrt{\gamma} \times \frac{\mathbf{w}^T \boldsymbol{\mu} - r_f}{\sqrt{\mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}}}.$$

- **Geometric Sharpe ratio:**

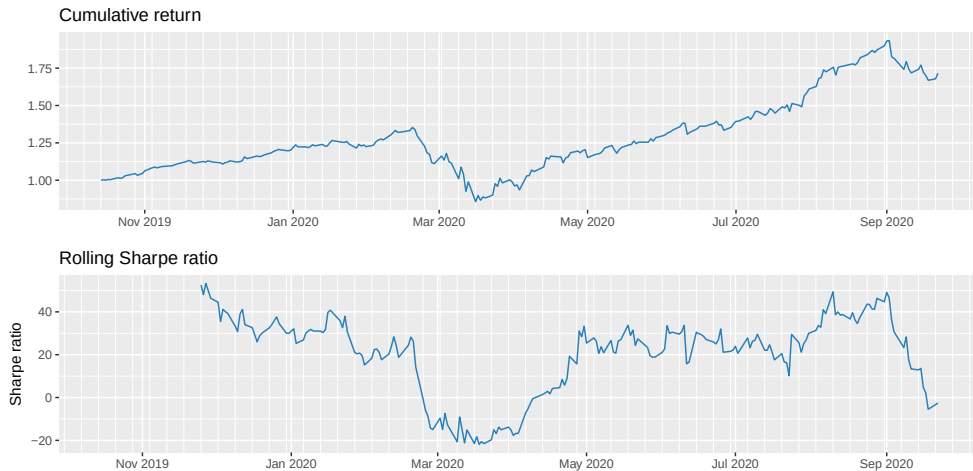
- Uses compounded annualized return in the numerator.
- Based on geometric mean of returns.

- **Rolling Sharpe ratio:**

- SR computed on a rolling-window basis.
- Useful for assessing performance over time.

Performance measures: Rolling Sharpe ratio

Cumulative returns and rolling annualized Sharpe ratio (overall SR of 1.65):



- **Information ratio (IR):**

- Compares portfolio return to a benchmark:

$$\text{IR} = \frac{\mathbb{E} [\mathbf{w}^T \mathbf{r}_t - r_t^b]}{\text{Std} [\mathbf{w}^T \mathbf{r}_t - r_t^b]},$$

- r_t^b : return of the benchmark.
- Measures excess return per unit of risk relative to the benchmark.

- **Downside risk and semivariance:**

- Variance measures deviation from mean return:

$$\text{Var} [\mathbf{w}^T \mathbf{r}_t] = \mathbb{E} \left[(\mathbf{w}^T \mathbf{r}_t - \mathbf{w}^T \boldsymbol{\mu})^2 \right].$$

- Downside risk focuses on underperformance relative to a reference level.

- **Semivariance:**

- Measures risk of return falling below the mean:

$$\text{SemiVar} [\mathbf{w}^T \mathbf{r}_t] = \mathbb{E} \left[\left((\mathbf{w}^T \boldsymbol{\mu} - \mathbf{w}^T \mathbf{r}_t)^+ \right)^2 \right],$$

- $(\cdot)^+$ avoids penalizing above-mean performance.

Performance measures: Downside risk and semivariance*

- **Semi-deviation/downside deviation:**

- Square root of semivariance, analogous to volatility.

- **Lower partial moments (LPM):**

- General downside risk measure:

$$\text{LPM}_\alpha(\tau) = \mathbb{E} \left[\left((\tau - \mathbf{w}^\top \mathbf{r}_t)^+ \right)^\alpha \right],$$

- τ : disaster level.
- α : reflects investor's risk attitude (risk-averse, neutral, risk-seeking).
- Can form various downside measures by choosing α and τ (e.g., semivariance with $\alpha = 2$ and $\tau = \mathbf{w}^\top \boldsymbol{\mu}$).

- **Gain-loss ratio (GLR):**

- Downside risk measure comparing positive and negative returns.
- Formula:

$$\text{GLR} = \frac{\mathbb{E} \left[R_t^{\text{portf}} \mid R_t^{\text{portf}} > 0 \right]}{-\mathbb{E} \left[R_t^{\text{portf}} \mid R_t^{\text{portf}} < 0 \right]}.$$

- Reflects the average gain to average loss ratio.
- Note: The denominator is made positive for a meaningful ratio by taking its absolute value.

- **Sortino Ratio:**

- Adjusts for downside risk instead of total volatility.
- Formula:

$$\text{SoR} = \frac{\mathbf{w}^T \boldsymbol{\mu} - r_f}{\sqrt{\text{SemiVar}[\mathbf{w}^T \mathbf{r}_t]}},$$

- Focuses on undesirable volatility (downside risk).
- More appropriate for investors concerned primarily with downside risk.

Performance measures: Value-at-risk (VaR)

- **Value-at-Risk (VaR):**

- VaR focuses on the tail of the distribution, indicating potential large losses.
- Measures maximum loss at a specified confidence level (e.g., 95%).
- Defined as the quantile of the loss distribution:

$$\text{VaR}_\alpha = \inf \{ \xi_0 \mid \Pr(\xi \leq \xi_0) \geq \alpha \},$$

where $\xi_t = -\mathbf{w}^\top \mathbf{r}_t$ represents the loss.

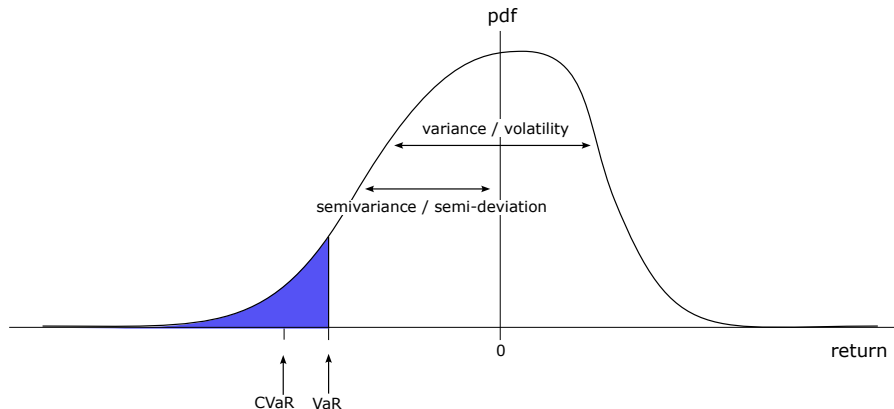
- α is the confidence level (e.g., $\alpha = 0.95$ for 95% confidence).

- **Drawbacks of VaR:**

- Does not quantify the shape of the tail beyond the VaR threshold.
- Cannot indicate the magnitude of losses exceeding the VaR.
- Not subadditive, challenging the principle of diversification (merging two portfolios can increase perceived risk).

Performance measures: Value-at-risk (VaR)

Illustration of distribution of returns and measures of risk:



- **Conditional Value-at-Risk (CVaR):**

- Expected value of the loss tail:

$$\text{CVaR}_\alpha = \mathbb{E}[\xi \mid \xi \geq \text{VaR}_\alpha].$$

- Also known as Expected Shortfall (ES).
- Takes into account the average value of losses exceeding the VaR.

- **Advantages of CVaR:**

- Provides insight into the magnitude of losses beyond the VaR threshold.
- Subadditive, aligning with the principle of diversification.

Performance measures: Drawdown

- **Drawdown (DD):**

- Measures decline from a historical peak in cumulative profit or NAV:

$$D_t = \text{HWM}_t - \text{NAV}_t,$$

- High Water Mark (HWM):

$$\text{HWM}_t = \max_{\tau \in [0, t]} \text{NAV}_\tau.$$

- **Normalized drawdown:**

- Relative decline from the peak:

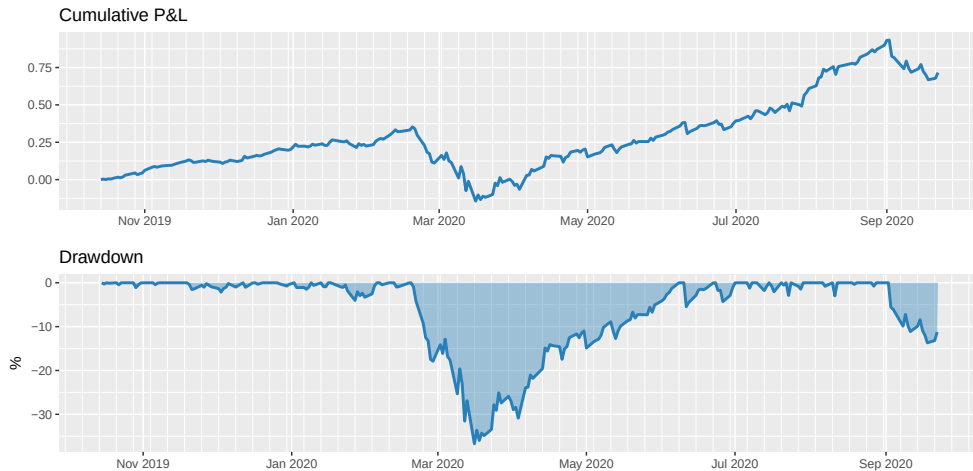
$$\bar{D}_t = \frac{\text{HWM}_t - \text{NAV}_t}{\text{HWM}_t}.$$

- **Drawdown as a path-dependent measure:**

- Depends on the sequence of returns, unlike other performance measures.

Performance measures: Drawdown

Cumulative P&L and corresponding drawdown of a portfolio:



Performance measures: Drawdown*

- **Maximum drawdown (Max-DD):**

- Largest observed drawdown:

$$\text{Max-DD} = \max_{1 \leq t \leq T} \bar{D}_t;$$

- **Average drawdown (Ave-DD):**

- Mean of drawdowns over time:

$$\text{Ave-DD} = \frac{1}{T} \sum_{1 \leq t \leq T} \bar{D}_t;$$

- **Conditional drawdown-at-risk (CDaR):**

- Expected drawdown in the worst cases:

$$\text{CDaR}_\alpha = \mathbb{E} [\bar{D}_t \mid \bar{D}_t \geq \text{VaR}_\alpha],$$

- VaR_α : Value at risk of drawdown at confidence level α .

Performance measures: Calmar ratio and Sterling ratio*

- **Calmar Ratio:**

- Adjusts Sharpe ratio by using maximum drawdown:

$$\text{Calmar ratio} = \frac{\mathbf{w}^T \boldsymbol{\mu} - r_f}{\text{Max-DD}}.$$

- Focuses on return per unit of maximum drawdown risk.

- **Sterling Ratio:**

- Similar to Calmar ratio, with an added risk measure:

$$\text{Sterling ratio} = \frac{\mathbf{w}^T \boldsymbol{\mu} - r_f}{\text{Max-DD} + 0.1}.$$

- Incorporates an excess risk measure to account for additional risk considerations.

Outline

- 1 Fundamentals
- 2 Portfolio constraints
- 3 Performance measures
- 4 Heuristic portfolios**
- 5 Risk-based portfolios
- 6 Summary

- **Heuristic portfolios overview:**

- Not mathematically derived but widely used in practice.
- Serve as benchmarks for portfolio performance.
- Include simple, rule-based strategies for asset allocation.

- **Examples:**

- buy&hold (B&H)
- global maximum return portfolio (GMRP)
- $1/N$ portfolio
- quintile portfolios

- **Basic reference:** (Palomar 2025, chap. 6).

- **Buy&Hold (B&H) Portfolio:**

- Simple investment strategy focusing on a single asset or a set of assets.
- Allocation:

$$\mathbf{w} = \mathbf{e}_i,$$

where $\mathbf{e}_i$ is an all-zero vector except for a one at the i th element.

- Lacks diversification in its single-asset form.
- Aimed at long-term investment without adjustments over short horizons.
- Based on the belief that the chosen asset(s) will appreciate over time.
- Asset selection can be guided by fundamental or technical analysis.

- **Global maximum return portfolio (GMRP):**

- Formulation:

$$\begin{aligned} & \underset{\mathbf{w}}{\text{maximize}} && \mathbf{w}^\top \boldsymbol{\mu} \\ & \text{subject to} && \mathbf{1}^\top \mathbf{w} = 1, \quad \mathbf{w} \geq \mathbf{0}, \end{aligned}$$

- Problem can be solved with an LP solver.
- However, in this case the solution is trivial: It allocates all the budget to the asset with the highest forecast return.

- **Cautions with GMRP:**

- Lacks diversification, potentially leading to poor performance.
- Past performance is not indicative of future results.
- Estimation of $\boldsymbol{\mu}$ is often noisy and unreliable.

- **Equally weighted portfolio (EWP) or $1/N$ portfolio:**

- Allocates capital equally across all N assets:

$$\mathbf{w} = \frac{1}{N} \times \mathbf{1},$$

where $\mathbf{1} \in \mathbb{R}^N$ is the all-one vector.

- Emphasizes diversification: “do not put all your eggs in the same basket.”

- **Historical context and performance:**

- Also known as “Talmudic rule” based on ancient recommendation.
- Has shown superior historical performance and inspired equally weighted ETFs.
- S&P 500 equally weighted indices developed by Standard & Poor.
- Claims of outperforming Markowitz’s mean-variance portfolio, though debated.

Heuristic portfolios: Quintile portfolios

- **Quintile portfolio:**

- Allocates equally to top 20% of assets based on performance ranking.
- Can be adjusted for different fractions (e.g., quartile, decile).
- Long-short version involves short-selling bottom-ranked assets.

- **Portfolio composition:**

- Assets ranked from top to bottom performers.
- Allocation:

$$\mathbf{w} = \frac{1}{N/5} \begin{bmatrix} 1 \\ \vdots \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad \left. \begin{array}{l} \text{ } \\ \text{ } \\ \text{ } \end{array} \right\} \begin{array}{l} 20\% \\ \\ 80\% \end{array}$$

Heuristic portfolios: Quintile portfolios

- **Ranking and factor investing:**

- Ranking based on various factors (Value, Low Size, Low Volatility, Momentum, Quality).
- Factors derived from characteristics like book-to-price ratio, market capitalization, standard deviation, past returns, return on equity.
- Empirical evidence of excess returns for these factors.

- **Public data ranking methods:**

- Expected performance:

$$\text{score} = \mu;$$

- Sharpe ratios:

$$\text{score} = \frac{\mu}{\sqrt{\text{diag}(\Sigma)}};$$

- Mean-variance ratios:

$$\text{score} = \frac{\mu}{\text{diag}(\Sigma)}.$$

Heuristic portfolios as robust portfolios

- **The $1/N$ and quintile portfolios as robust portfolios:**

- In principle, they are just heuristic portfolios.
- Nevertheless, they can be formally derived as portfolios robust to estimation errors in μ .

- **Robust formulation of GMRP:**

- Maximizes the worst-case return considering uncertainty in μ :

$$\begin{aligned} & \underset{\mathbf{w}}{\text{maximize}} && \min_{\mu \in \mathcal{S}} \mathbf{w}^T \mu \\ & \text{subject to} && \mathbf{1}^T \mathbf{w} = 1, \quad \mathbf{w} \geq \mathbf{0}, \end{aligned}$$

- Uncertainty set $\mathcal{S}$:

$$\mathcal{S} = \{\hat{\mu} + \mathbf{e} \mid \|\mathbf{e}\|_1 \leq \epsilon\},$$

where ϵ is the size of the uncertainty region.

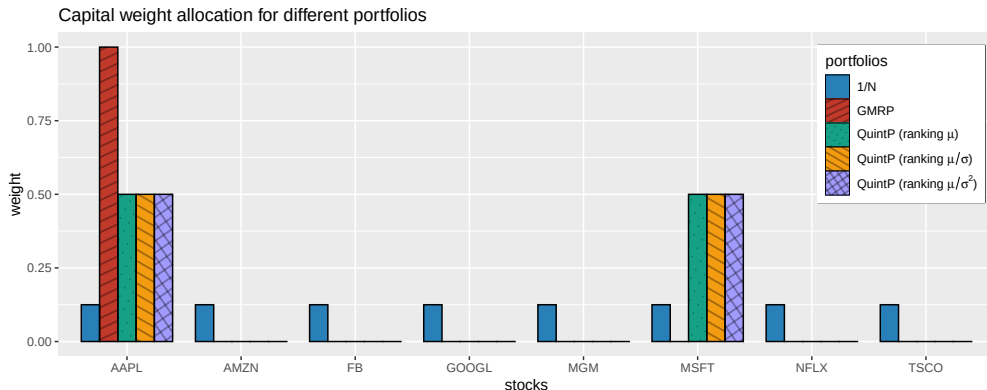
- **Connection to quintile and $1/N$ portfolios:**

- Quintile portfolio emerges as a solution for a specific ϵ .
- $1/N$ portfolio arises as a solution for a sufficiently large ϵ .
- Explains the practical performance of these portfolios due to their robustness.
- Further details in robust portfolios chapter: (Palomar 2025, chap. 14).

Heuristic portfolios: Numerical experiments

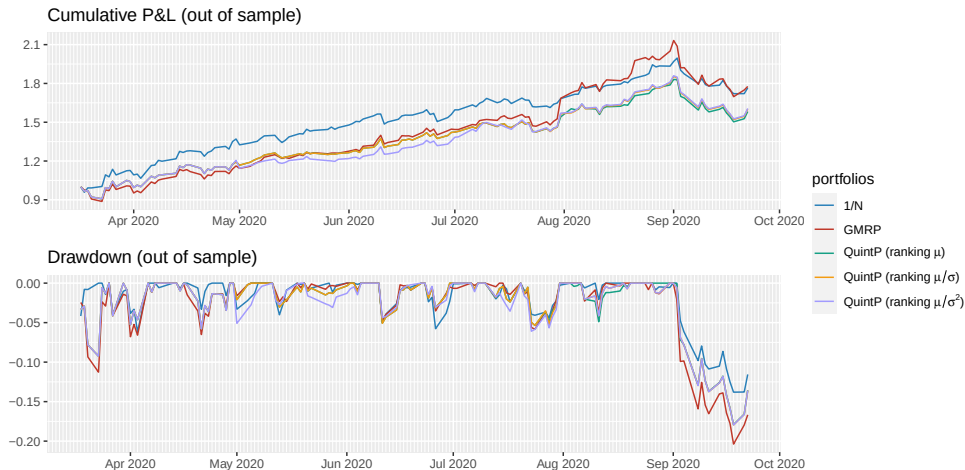
- **Comparison of heuristic portfolios:**

- Portfolios: GMRP, $1/N$, and quintile portfolios (ranked by μ , μ/σ , μ/σ^2).
- Data: Eight S&P 500 stocks, 2019-2020.
- Backtest: 70% training data, 30% test data.



Heuristic portfolios: Numerical experiments

Backtest cumulative P&L and drawdown:



- **Backtest performance analysis:**

- GMRP shows high returns but increased volatility and drawdown.
- $1/N$ portfolio exhibits the best drawdown performance, indicating good diversification.

- **Considerations for realistic backtesting:**

- Multiple rolling-window backtests recommended for robust assessment.
- Chapter on backtesting provides detailed methodologies.

Outline

- 1 Fundamentals
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- 3 Performance measures
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- 6 Summary

- **Risk-based portfolios overview:**

- Focus on risk considerations, utilizing the covariance matrix Σ .
- Ignore return forecasts μ due to their unreliable estimation.
- Aim to achieve diversification and risk management without relying on expected returns.

- **Characteristics:**

- Based on the premise that risk parameters (Σ) are more reliably estimated than returns (μ).
- Seek to optimize portfolio construction through risk minimization or distribution.

- **Significance:**

- Addresses the challenge of high uncertainty in return forecasts.
- Provides a framework for portfolio construction when return predictions are deemed too speculative or volatile.

- **Examples:**

- Global minimum variance portfolio (GMVP)
- Inverse volatility portfolio (IVolP)
- Inverse variance portfolio (IVarP)
- Risk parity portfolio (RPP)
- Most diversified portfolio (MDP)
- Maximum decorrelation portfolio (MDCP)

- **References:**

- (Ardia et al. 2017) for detailed exploration of risk-based portfolios.
- (Chopra and Ziemba 1993) highlighting the issues with estimating μ in practice.
- (Palomar 2025, chap. 6) for a detailed coverage of these slides.

Risk-based portfolios: Global minimum variance portfolio (GMVP)

- **Global minimum variance portfolio (GMVP):**

- Objective: Minimize portfolio variance.
- Formulation:

$$\begin{aligned} & \underset{\mathbf{w}}{\text{minimize}} && \mathbf{w}^T \Sigma \mathbf{w} \\ & \text{subject to} && \mathbf{1}^T \mathbf{w} = 1, \quad \mathbf{w} \geq \mathbf{0}. \end{aligned}$$

- Convex quadratic program (QP) solvable with QP solvers.
- Without shorting constraints it has the closed-form solution:

$$\mathbf{w} = \frac{1}{\mathbf{1}^T \Sigma^{-1} \mathbf{1}} \Sigma^{-1} \mathbf{1}.$$

- **Significance:**

- Utilized for its simplicity and as a benchmark in academic research.
- Focuses on risk minimization through variance reduction.
- Often used to evaluate covariance matrix estimators, excluding μ estimation effects.

Risk-based portfolios: Inverse volatility portfolio (IVolP)

- **Inverse volatility portfolio (IVolP):**

- Aims to control risk by inversely weighting assets based on their volatility.
- Defined as:

$$\mathbf{w} = \frac{\boldsymbol{\sigma}^{-1}}{\mathbf{1}^\top \boldsymbol{\sigma}^{-1}},$$

where $\boldsymbol{\sigma}^2 = \text{diag}(\boldsymbol{\Sigma})$ and $\boldsymbol{\sigma}$ are the assets' volatilities.

- **Characteristics:**

- Lower weight to high volatility assets, higher weight to low volatility assets.
- Also known as *equal volatility portfolio*.
- Ensures weighted constituent assets have equal volatility contribution to the portfolio.

- **Comparison with GMVP:**

- The GMVP with $\boldsymbol{\Sigma}$ diagonal leads to the inverse variance portfolio (IVarP) (resembling the IVolP):

$$\mathbf{w} = \frac{\boldsymbol{\sigma}^{-2}}{\mathbf{1}^\top \boldsymbol{\sigma}^{-2}},$$

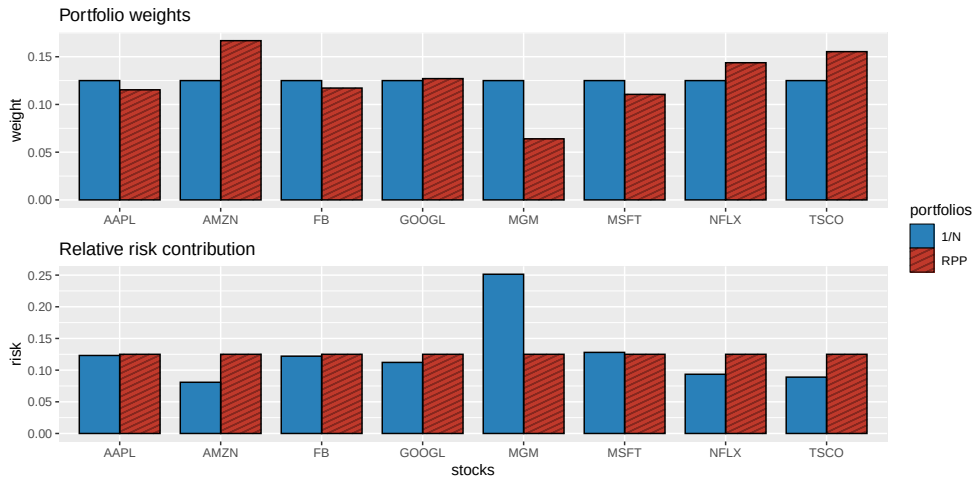
- Both strategies allocate inversely based on volatility, but the IVarP squares the volatilities (variance).

Risk-based portfolios: Risk parity portfolio (RPP)

- **Risk parity portfolio (RPP) or equal risk portfolio (ERP):**
 - Aims to equalize risk contributions from all assets to the total portfolio risk.
 - Considers both asset volatilities and correlations.
 - Detailed exploration in Chapter @ref(RPP).
- **Comparison with IVolP:**
 - IVolP focuses on inversely weighting assets based on their volatilities, ignoring correlations.
 - RPP advances this concept by incorporating asset correlations, offering a more comprehensive risk management approach.
- **Objective:**
 - Distribute risk evenly across assets, unlike $1/N$ portfolio which equalizes capital allocation.
 - Ensures no single asset disproportionately affects the portfolio's overall risk.
- **Significance:**
 - Recognizes and addresses the impact of asset correlations on portfolio risk.
 - Provides a balanced approach to risk management, suitable for diversified investment strategies.

Risk-based portfolios: Risk parity portfolio (RPP)

From dollar diversification ($1/N$ portfolio) to risk diversification (RPP):



Risk-based portfolios: Most diversified portfolio (MDP)*

- **Most diversified portfolio (MDP):**

- Utilizes the diversification ratio (DR) as a proxy for risk efficiency.
- Defined as:

$$DR = \frac{\mathbf{w}^T \boldsymbol{\sigma}}{\sqrt{\mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}}}.$$

- Long-only portfolios have $DR \geq 1$.

- **MDP formulation:** Maximizes DR

$$\begin{array}{ll} \underset{\mathbf{w}}{\text{minimize}} & \frac{\mathbf{w}^T \boldsymbol{\sigma}}{\sqrt{\mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}}} \\ \text{subject to} & \mathbf{1}^T \mathbf{w} = 1, \quad \mathbf{w} \geq \mathbf{0}. \end{array}$$

- **Significance:**

- MDP aims to maximize diversification benefits by considering asset volatilities and correlations.
- Offers a portfolio construction approach that aligns with the concept of risk efficiency in markets.

Risk-based portfolios: Maximum decorrelation portfolio (MDCP)*

- **Maximum decorrelation portfolio (MDCP):**

- Minimizes the portfolio correlation:

$$\begin{aligned} & \underset{\mathbf{w}}{\text{minimize}} && \mathbf{w}^T \mathbf{C} \mathbf{w} \\ & \text{subject to} && \mathbf{1}^T \mathbf{w} = 1, \quad \mathbf{w} \geq \mathbf{0}. \end{aligned}$$

- Correlation matrix $\mathbf{C}$ represents covariance of normalized assets with unit volatility.

- **Relation to GMVP and MDP:**

- MDCP is akin to GMVP with correlation matrix instead of covariance matrix.
- MDCP weights, when adjusted for asset volatilities, yield MDP weights.

- **Properties of MDCP:**

- Maximizes the Diversification Ratio (DR) when assets have equal volatility.
- Maximizes the Sharpe Ratio (SR) when assets have equal risks and returns.

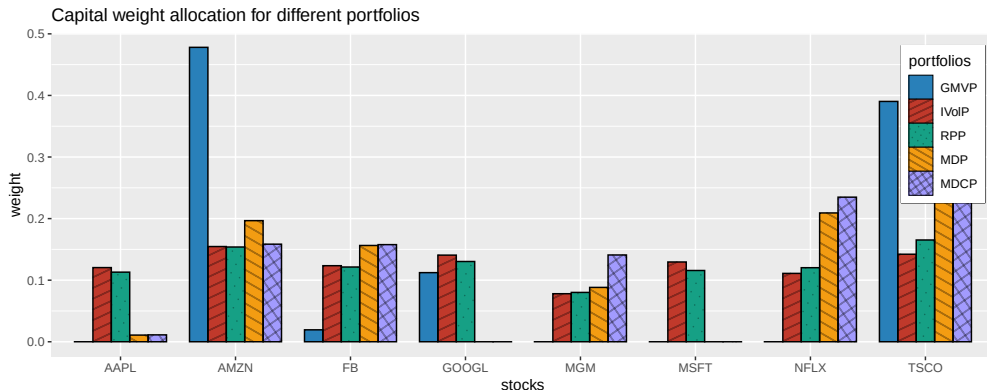
- **Significance:**

- MDCP focuses on reducing portfolio correlation, enhancing diversification.
- Offers an alternative risk management strategy by considering asset correlations.

Risk-based portfolios: : Numerical experiments

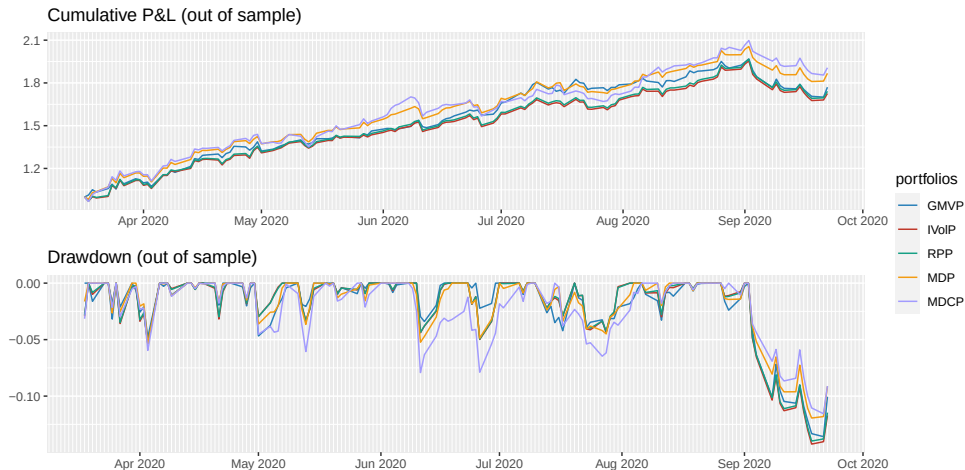
- **Comparison of risk-based portfolios:**

- Portfolios: GMVP, IVolP, RPP, MDP, and MDCP.
- Data: Eight S&P 500 stocks, 2019-2020.



Risk-based portfolios: Numerical experiments

Backtest cumulative P&L and drawdown:



- **Backtest performance analysis:**

- Emphasizes the effectiveness of risk-based portfolios in minimizing drawdowns compared to heuristic portfolios.
- Showcases the impact of different risk minimization strategies on portfolio performance.

- **Observations:**

- Risk-based portfolios generally exhibit smaller drawdowns, highlighting their focus on risk minimization.
- The allocation and performance of these portfolios demonstrate the practical implications of different risk management approaches in portfolio construction.

Outline

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- 2 Portfolio constraints
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Summary

- A portfolio of N assets is represented by vector $\mathbf{w}$, with return $\mathbf{w}^T \mathbf{r}$ where $\mathbf{r}$ is the asset returns vector.
- Portfolios require periodic rebalancing, which incurs transaction costs that reduce returns.
- Portfolios must meet various constraints, including regulatory, broker-imposed, and investor-specific ones, most of which are convex for easy optimization, except sparsity control.
- Portfolio performance is assessed using metrics like Sharpe ratio and CVaR, and incorporating these into optimization creates complex formulations.
- Simple heuristic portfolios like the $1/N$ and quintile portfolios are easy to compute and historically effective.
- Risk-based portfolios aim to minimize return variability without market trend speculation, offering ease of computation and practical robustness.

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- Palomar, D. P. 2025. *Portfolio Optimization: Theory and Application*. Cambridge University Press.